

Spencer Teetaert, PhD Student

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🌐 <https://spencerteetaert.github.io/>

Education

- 2023 – present 📌 **Ph.D. Student in Robotics Engineering, University of Toronto** under the supervision of Dr. Timothy D. Barfoot and Dr. Jessica Burgner-Kahrs. Research focus on stochastic state estimation for continuum robots.
- 2018 – 2023 📌 **B.A.Sc. Engineering Science, University of Toronto** in Robotics. GPA: 3.77 / 4.0
Thesis: *Towards Learning-Based Control for a Tendon-Driven Parallel Continuum Robot.*

Research Contributions

Papers

- 1 **S. Teetaert**, G. Caroleo, M. Pontin, *et al.*, “Continuum robot localization using distributed time-of-flight sensors,” 2026, Submitted to *Robotics Science and Systems (RSS)*. arXiv: 2602.07209.
- 2 **S. Teetaert**, S. Lilge, J. Burgner-Kahrs, and T. D. Barfoot, “A sliding-window filter for online continuous-time continuum robot state estimation,” in *2026 IEEE 9th International Conference on Soft Robotics (RoboSoft)*, 2026, forthcoming.
- 3 **S. Teetaert**, S. Lilge, J. Burgner-Kahrs, and T. D. Barfoot, “A stochastic framework for continuous-time state estimation of continuum robots,” 2025, Submitted to *IEEE Transactions on Robotics*. arXiv: 2510.01381.
- 4 **S. Teetaert**, W. Zhao, A. Loquercio, *et al.*, “Advancing reproducibility, benchmarks, and education with remote sim2real: Remote simulation to real robot hardware,” *IEEE Robotics & Automation Magazine*, vol. 32, no. 1, pp. 117–123, 2025. 📄 DOI: 10.1109/MRA.2025.3527291.
- 5 **S. Teetaert**, W. Zhao, N. Xinyuan, *et al.*, “A remote sim2real aerial competition: Fostering reproducibility and solutions’ diversity in robotics challenges,” 2023. arXiv: 2308.16743.

Workshops and Posters

- 1 **S. Teetaert***, E. Walsh*, E. Diller, T. D. Barfoot, and J. Burgner-Kahrs, *Sliding sensors: Configurable confidence in state estimation for continuum robots*, Submitted to *2026 IEEE 9th International Conference on Soft Robotics (RoboSoft)*, 2026.
- 2 **S. Teetaert**, S. Lilge, J. Burgner-Kahrs, and T. D. Barfoot, *Space-time continuum: Continuous shape and time state estimation for flexible robots*, 40th Anniversary of the IEEE International Conference on Robotics and Automation (ICRA40), 2024.
- 3 **S. Teetaert**, *Accessible sim2real for benchmarking and hands-on education*, Workshop on Benchmarking, Reproducibility, and Open-Source Code in Controls, IEEE Conference on Decision and Control, 2023.

Other

- 1 Co-organizer, “Safe robot learning competition,”
<https://www.dynsyslab.org/iros-2022-safe-robot-learning-competition/>, 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).

Employment History

- 2021 – 2022  **Research Engineer Coop**, Human Machine Interaction Lab, Huawei Canada.
Developed advanced machine learning computer vision models for eye-gaze and gesture tracking with applications to consumer electronic devices.
- 2019 – 2020  **Industrial Engineering Coop**, Hylife Foods
Launch an automation overhaul project for a production plant of 2,000+ workers. Built relationships with third party robotics partners and aided in design projects. Lead a cutting blade optimization project.

TA Positions

- 2024 – present  **ROB498**: Robotics Capstone Design, University of Toronto.
- 2023  **ROB301**: Introduction to Robotics, University of Toronto.

Skills

- Coding  C++, Python, MATLAB, L^AT_EX
- Misc.  3D CAD Design (Autodesk Inventor, Fusion360, SolidWorks)
-  Basic Machining, George Brown College

Miscellaneous Experience

Awards and Achievements

- 2025-26  **University of Toronto Robotics Institute Fellow**
-  **Queen Elizabeth II Graduate Scholarship in Science and Technology**
- 2024-25  **Ontario Graduate Scholarship**
- 2022  **NSERC Undergraduate Student Research Award**
- 2018  **Faculty of Applied Science & Engineering Admission Scholarship**
-  **Dr. Dale Iwanoczko Memorial Scholarship**

Extra-Curriculars and Volunteer

- 2025–present  **University of Toronto Undergraduate Research Students' Association**
• *Graduate Student Mentor*
- 2020–2023  **University of Toronto Robotics Association**
• *President*
• *Project Manager, Autonomous Rover Team*
• *Computer Vision Designer, Autonomous Rover Team*
- 2019–2023  **University of Toronto Engineering Orientation**
• *Subcommittee Chair, Photography*
• *Subcommittee Chair, Retreat*
• *Group Leader*